Jasper Sands

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RESEARCH INTERESTS

Quantum error correction and decoding · quantum algorithms and query complexity · fault-tolerant resource estimation · machine learning for quantum-device characterization and control.

EDUCATION

Columbia University, New York, NY — Sep 2025 – May 2027

Master of Science in Computer Science, Quantum Computing Track. GPA 3.95/4.0.

Thesis: *Automation of Silicon Spin Qubit Measurements Using Machine Learning.* Advised by Dr. Nard Dumoulin Stuyck at Diraq and UNSW Sydney, with Dr. Henry Yuen at Columbia.

Coursework: Quantum Error Correction · Quantum Engineering · Quantum Simulation & Computing Lab · Quantum Optimization & Machine Learning · Neural Networks & Deep Learning · Machine Learning for Functional Genomics.

Washington University in St. Louis, St. Louis, MO — Aug 2021 – May 2025

Bachelor of Science in Computer Science and Economics; Minor in Political Science. GPA 3.81/4.0.

Coursework: Machine Learning · Analysis of Algorithms · Algorithms for Biosequence Comparison.

TECHNICAL REPORTS

Jasper Sands. *Factoring and Chemistry on a Simulated Fault-Tolerant Quantum Computer.* 2026.

Jasper Sands. *A Surface-Code Simulator and Decoder, Checked against Stim, PyMatching and Google's Hardware.* 2026.

Kate Bonner and Jasper Sands. *Noise-Aware Ansatz Design for Digitized Adiabatic Quantum Computing.* 2025.

Mike Azure and Jasper Sands. *Practical LLM-Based Lossless Compression: An Empirical Study.* 2025.

John C. Papciak and Jasper Sands. *Multimodal Hyperbolic Embeddings for AMR Detection and Taxonomic Modeling in Metagenomic Contigs.* 2025.

RESEARCH EXPERIENCE

Diraq and UNSW Sydney · *MS Thesis Research* — Sep 2026 – May 2027

- Automating silicon spin-qubit characterization under Dr. Nard Dumoulin Stuyck, targeting manual tuning and interpretation of charge measurements that bottleneck semiconductor quantum-hardware development.
- Building pipelines that acquire charge-measurement data from cryogenic probing and classify charge transitions and device states with machine learning, benchmarked against expert interpretation.

Cidon Systems Research Lab, Columbia University · *Graduate Researcher* — Sep 2025 – Jun 2026

- Built phishing and spam detection with Dr. Asaf Cidon and Barracuda from LLM embeddings and email telemetry.

Complex Resilient Intelligent Systems Lab, Columbia University · *Graduate Researcher* — May 2025 – Dec 2025

- Reduced hallucination in generative protein modeling with Dr. Venkat Venkatasubramanian by grounding outputs in ontology and molecular-graph data.

Foster Care Policy Database, Washington University in St. Louis · *Undergraduate Researcher* — Sep 2024 – May 2025

- Fine-tuned LLaMA-3 under Dr. Ian Fillmore for question answering over 50k foster-care policy documents, with human-in-the-loop feedback for cross-state policy analysis.

SELECTED PROJECTS

stabilizer-qec: surface-code simulator and decoder · *Rust, Python, WebAssembly* — PyPI, DOI

- Stabilizer simulator and decoders (sparse-blossom and correlated matching, BP+OSD, Union-Find) for surface, XZZX and IBM gross codes and lattice surgery; agrees with Stim, PyMatching and *ldpc* edge for edge and shot for shot.
- Rebuilt all of Google's Willow and Sycamore memory experiments (27.5M shots) from raw measurements bit for bit and fitted Λ as Google does; a windowed decoder keeps up with a million-round d = 5 stream on 4 cores.

ft-algorithms: algorithms on a simulated fault-tolerant quantum computer · *Rust, Python* PyPI, DOI

- Runs Shor's algorithm (N up to 143) and phase estimation for H_2, HeH^+ and LiH on a lattice-surgery machine whose logical error channels are fitted to 316 circuit-level surface-code experiments, with magic-state factories.
- Its resource estimator reproduces Gidney's 2025 RSA-2048 distance (d = 25) under his noise projection and Lee et al.'s FeMoco d = 31; under the measured noise, one RSA-2048 shot needs d = 27–29.

Q-Search: proof-gated quantum algorithm research and verification engine

- Searches for structural quantum advantage on non-abelian hidden subgroup, dihedral coset and code equivalence.
- Formal proof gates test each proposed mechanism against 1,200+ classical baselines across 720+ theorem modules, indexing negative results so dead ends are not re-proposed.

WhatTheDuck: quantum amplitude estimation for Value at Risk · *CUDA-Q, C++* · MIT iQuHACK 2026

- Wrote GPU-accelerated state preparation, threshold oracle, and bisection search for Iterative Quantum Amplitude Estimation; measured query scaling matched the $O(1/\varepsilon)$ bound against an $O(1/\varepsilon^2)$ classical baseline.

TEACHING EXPERIENCE

Introduction to Quantum Computing, Columbia University · *Teaching Assistant* Jan 2026 – May 2026

Parallel and Concurrent Programming, Washington University · *Teaching Assistant* Jan 2025 – May 2025

Reverse Engineering and Malware Analysis, Washington University · *Teaching Assistant* Aug 2024 – Dec 2024

Taught a lecture in the quantum computing and malware analysis courses, plus office hours and grading.

INDUSTRY EXPERIENCE

Smack Technologies · *AI Intern* May 2026 – Aug 2026

- Built physics and sensor simulations for AI pipelines, using knowledge-graph state to expand corpus generation.
- Benchmarked open-weight models across task performance, inference speed, and concurrency, raising throughput 64×.
- Validated simulator outputs against independently selected reference cases, bounding error to 0.55%, in a report that regenerates when the physics changes.

Highnote · *Cybersecurity Engineer Intern* May 2024 – Jul 2024

- Built the company-wide SIEM for a card-issuing fintech: AWS, GCP and Datadog telemetry in one Elasticsearch cluster (1 TB/day), with 100+ rule-based and ML detections.

Mindtrip · *Software Engineer Intern* May 2023 – Jul 2023

- Built Retool admin and secure-query tooling replacing direct production database writes, saving ~250 hours a month.

COMPETITIONS

- **Overall Winner** & **NVIDIA Ecosystem Award**, MIT iQuHACK 2026, State Street/Classiq Challenge, qualified for the NYU Abu Dhabi Quantum and AI Hackathon for Social Good 2027.
- **Audience Favorite**, Harmoniqs Quantum Design Hackathon 2026.
- **Runner-Up**, Qualcomm Snapdragon Hackathon 2026.

TECHNICAL SKILLS

Quantum tools: Qiskit, Qiskit Runtime, Cirq, PennyLane, CUDA-Q, Stim, PyMatching, QuTiP.

Quantum algorithms: Hamiltonian simulation, quantum phase estimation, amplitude amplification and estimation, QSVT, block encoding, linear combination of unitaries, quantum walks, VQE, QAOA, Trotter decomposition.

Quantum error correction: Rotated and XZZX surface codes, stabilizer formalism, MWPM and Union-Find decoding, finite-size scaling, threshold estimation, noise modeling, lattice surgery, resource estimation.

Programming: Python, C++, Rust, C, MATLAB, JavaScript/TypeScript, Go, Java.

ML & systems: PyTorch, scikit-learn, Hugging Face, CUDA, NumPy, SciPy, Linux, Git, Docker, AWS, GCP.

LEADERSHIP AND SERVICE

- **Graduate Lead**, Columbia Quantum Algorithms Reading Group, Sep 2025 – Jun 2026. Led weekly sessions covering all 33 chapters of Andrew Childs' quantum algorithms notes, paired with simulation checks.
- **Founding team member**, Columbia Autonomous Racing Club. Coordinated sponsor outreach and track testing.
- **Volunteer Tutor**, St. Louis Tutor Me Program. K-12 mathematics and literacy in underserved districts.
- **Team Captain**, WashU Gymnastics Club.